

Why tomorrow's aircraft are still years away from takeoff

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By Tommaso Lecca · Jul 24, 2026, 4:56 PM · [View in your browser](#)

FARNBOROUGH, England — Futuristic aircraft concepts are a hallmark of every airshow — and the U.K.'s biennial Farnborough International Airshow is no exception. Just don't ask when you'll be able to fly in one.

From hybrid-electric airliners to hydrogen aircraft and flying taxis, manufacturers are making genuine technical progress. The biggest hurdles are no longer proving the technology can fly: They're certifying aircraft, building entirely new fueling and charging infrastructure, and persuading investors to keep funding projects that can be decades away from commercial service.

GE Aerospace announced Monday "the world's first high-altitude hybrid electric flight," completed in partnership with NASA, Boeing and aircraft developer BETA Technologies.

The modified aircraft, which succeeded in flying at the cruising altitude of commercial airliners, was equipped with a conventional CT7 engine on one side and a hybrid-electric powertrain on the other, said Christine Andrews, hybrid electric leader at GE Aerospace.

But Mohamed Ali, CEO of GE Aerospace's commercial engines business, cautioned that the technology remains some distance from powering entire planes transporting passengers. "We're not in a design phase yet. We are still in the technology maturation development phase for what the next generation applications are going to be," Ali told reporters at Farnborough.

The plane — a 34-seat Saab 340B — completed a transatlantic crossing. But while it departed New York on June 26, it only arrived in Farnborough on July 14.

Over the Atlantic, the hybrid-electric system operated alone for just over two hours, with the conventional engine providing propulsion for the remainder of the journey, which included multiple stopovers. Regulatory approvals also delayed the trip, Andrews said.

Even after successful demonstrations, the engine, especially with alternative propulsion and for air transport use, requires strict regulatory approval, while certifying an entire new large aircraft typically takes five to seven years, according to the EU Aviation Safety Agency .

See you in a few decades

That long timeline helps explain why Airbus is already planning its next "clean-sheet" aircraft — built from scratch to incorporate new tech, rather than based on existing models — more than a decade

before they are expected to enter service.

“We are preparing for the next generation of single-aisle aircraft, which we aim to introduce in the second half of the 2030s,” said Bruno Fichet, Airbus’ head of future programs. This new plane, with a conventional power source, will improve the technology and efficiency of existing aircraft.

When alternative propulsion systems are involved, manufacturers also need the infrastructure to support them.

Airbus launched its ZEROe program in 2020 with the ambition of developing a hydrogen-powered aircraft that would enter into service in 2035 . It has since pushed that timeline into the 2040s .

“The hydrogen ecosystem is taking more time than expected,” Fichet said, referring to the lack of availability of hydrogen at airports in sufficient quantities and at competitive prices.

The company has also abandoned plans to burn hydrogen directly in engines in favor of hydrogen fuel cells that generate electricity to power electric motors. Earlier this month, Airbus announced a joint venture with Germany’s MTU Aero Engines to develop and commercialize that technology.

Airbus now plans to develop a hydrogen-powered aircraft capable of carrying around 100 passengers over 1,000 nautical miles — roughly the distance between Madrid and Berlin.

Flying taxis are (almost) back

Flying taxi developers face a different obstacle: convincing investors to stay patient while certification drags on.

Bristol-based Vertical Aerospace said Monday it had completed “the first public electric vertical take-off and landing (eVTOL) flight at Farnborough International Airshow.”

The company is among the leading developers of air taxis — small electric aircraft designed to transport passengers over congested cities or serve remote communities.

But the aircraft that flew at the show is still a prototype. The production version is still awaiting certification from civil aviation authorities — a milestone Vertical initially expected to reach in 2025 but has now pushed back to 2029.

Despite the delay, the U.K. government awarded Vertical and BETA Technologies “up to £3.2 million funding to demonstrate how their next-gen aircraft can fit into commercial airport operations,” said the Department for Transport.

Vertical is also looking for investors “focused on the long-term value” rather than those who aim “to make money for the next three months and [then] ... buy AI stock and sell eVTOL,” said Vertical Chief Engineer David King.

Promises meet reality

The sector has learned that lesson the hard way.

Only a few years ago, air taxi developers were promising to carry passengers during the Paris 2024 Olympic Games .

“Have you seen them?” asked Augusto Viansson Ponte, London director at Alton Aviation Consultancy, with a hint of sarcasm.

Some promises made in recent years “were completely unsustainable in terms of speed to market from a certification perspective,” Viansson Ponte said.

Even aviation giants such as Airbus and Boeing “don’t necessarily get it right every time” when navigating certification, he added.

But certification is only part of the challenge. Many early business plans also relied on operating costs falling to unrealistic levels “that were, frankly, like dreamland.”

The industry’s dependence on outside investment can encourage companies to promise overly ambitious timelines.

From their perspective, Viansson Ponte said, “I need a fantastic project in order to be able to attract investment. But I need the investment in order to create a fantastic product.”

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